Planck’s Hypothesis

What was Planck's quantum hypothesis?

The quantum hypothesis, first suggested by Max Planck (1858–1947) in 1900, postulates that light energy can only be emitted and absorbed in discrete bundles called quanta. Planck came up with the idea when attempting to explain blackbody radiation, work that provided the foundation for his quantum theory.

State the postulates of Planck's quantum theory.

According to Planck’s quantum theory,

1.Different atoms and molecules can emit or absorb energy in discrete quantities only. The smallest amount of energy that can be emitted or absorbed in the form of electromagnetic radiation is known as quantum.

2.The energy of the radiation absorbed or emitted is directly proportional to the frequency of the radiation.

In 1900, Max Planck had studied radiations emitted by **hot black bodies** and presented the quantum theory of radiations explaining **electromagnetic radiation and energy.**

Postulates of the theory are:

1. Energy radiated by hot bodies is **not continuous**, rather it is emitted in the form of small “**energy packets**” or “**energy pulse**” known as **quanta** (singular is quantum). In the case of light , the quantum of energy is called a **photon**.

2. Each quantum has a **definite amount of energy**. This energy of the quantum is directly proportional to frequency of radiations.

$$E \propto \nu$$

Or $$E = h\nu$$

How did Einstein use Planck's hypothesis?

Planck announced his findings in 1900, and in 1905, Albert Einstein used Planck's quantum theory to describe the particle properties of light. Einstein demonstrated that electromagnetic radiation, including light, has the characteristics of both a wave and, consistent with Planck's theory, a particle

What is the significance of Planck's hypothesis?

It explains the quantum nature of the energy of electromagnetic waves. Planck's quantum theory deals with phenomena such as the photoelectric effect and the nature of radiated emission which were not explained by the laws of classical mechanics.

What are the applications of Planck's theory?

Planck's quantum theory is the fundamental theory of quantum mechanics. So, it has applications in all those fields where quantum mechanics is being used. It has applications in electrical appliances, the medical field, quantum computing, lasers, quantum cryptography etc.

Quantization of energy

What is meant by quantization of energy?

An electron can radiate or absorb energy as radiations only in limited amounts or bundles called quanta. This is known as the quantization of energy.

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WHAT IS QUANTA?

Quanta can be defined as the minimum quantity of energy that can either be gained or lost by an electron. Hence, if you see the energy levels of an electron, you can see that an atom needs to gain or lose energy to reach the next level or fall to the level. Absorption of heat or light helps the electrons gain energy to go to the next energy level or fall to lower energy levels.

Einstein used idea of Planck to explain his photoelectric effect. He confirmed that light consists of discrete units of energy known as photons carrying energy E=hv. Thus photoelectric effect confirmed the particle nature of light. Wave nature of light is already known due to phenomena like interference, diffraction, polarization etc. This dual nature of light is known as wave-particle duality.

De Broglie wave-particle duality hypothesis, Heisenberg‟s Uncertainty Principle and Schrodinger‟s equation provide base on which quantum mechanics is built.

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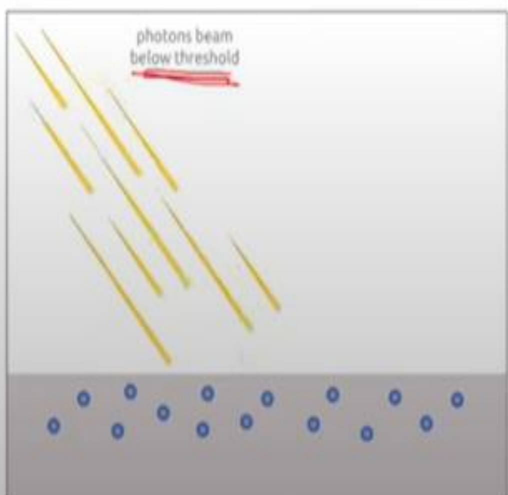
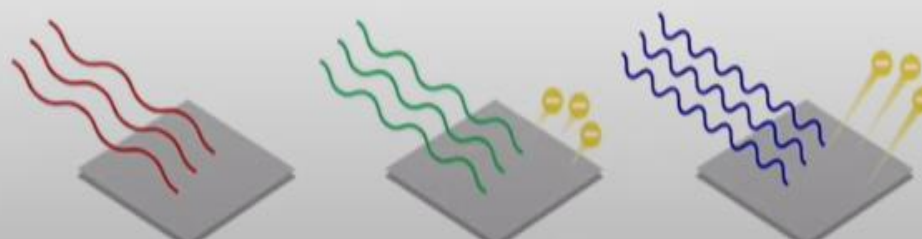
- He confirmed that light consists of discrete units of energy known as photons carrying energy - $E = h\nu$.
- **This dual nature of light is known as wave-particle duality.**

- The photoelectric effect is the emission of electrons or other free carriers when light hits a material
- Electrons emitted in this manner can be called photoelectrons

100		25		75
Energy Supplied	=	Work Function	+	Kinetic Energy

- Work function or threshold energy is the energy required by the electron to just begin emission





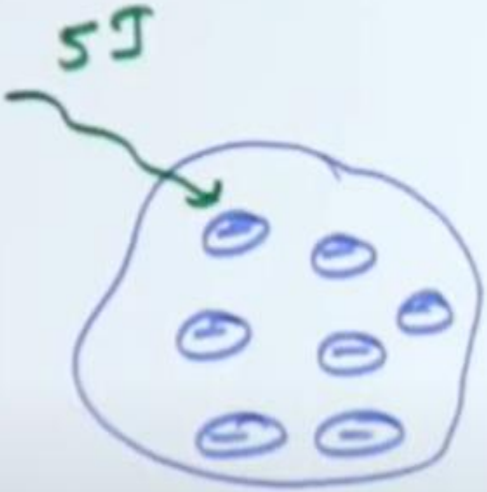
Einstein's Photoelectric Equation

Photoelectric

light → Photo

e^- → electric

electric current



$5J = 3J + 2J$

$2J$ → kinetic energy to e^-

$5J$ → energy of inci photon E

$3J$ → energy use to remove e^- from surface of metal.

[Work function

$$\boxed{E = W + KE}$$

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$$\boxed{h\nu = h\nu_0 + \frac{1}{2}mv^2} \quad (1)$$

→ This eqn is known as Einstein's photoelectric eqn.

$$h\nu - h\nu_0 = \frac{1}{2}mv^2$$

$$\boxed{h(\nu - \nu_0) = \frac{1}{2}mv^2} \quad (2)$$

definition

1. Threshold frequency →: Minimum freq required to remove e^- from the surface of metal called as threshold frequency (ν_0)

2) Threshold wavelength (λ_0) → Maximum wavelength required to remove e^- from the surface of metal called as threshold wavelength.

3) Workfunction → Energy required to remove the electron from surface of metal. (W or ϕ)

$$\phi = h\nu_0 = \frac{hc}{\lambda}$$

$E = W + KE$ W = workfunct^n

$h\nu = h\nu_0 + KE$.

$(KE)_{max} = h\nu - h\nu_0$

$= h(\nu - \nu_0)$

$$\boxed{(KE_{max}) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)}$$

$\nu \leftarrow \lambda$

$\nu = \frac{c}{\lambda}$

$\nu_0 \rightarrow$

$\lambda_0 -$

We know

$KE = eV_0$

$(KE_{max}) = eV_0$

Vo- stopping potential, is the negative potential, for which no emission of electrons.

$$\boxed{eV_0 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)}$$

Condition for photoemission.

1) $\nu \geq \nu_0$

2) $\lambda \leq \lambda_0$

if incident Photon of wavelengths 3500 A° and 4500 A°, and threshold wavelength is 4000 A° then

$\lambda_1 = 3500\ A^\circ$

$\lambda_2 = 4500\ A^\circ$

$\lambda_0 = 4000\ A^\circ$

$$\lambda < \lambda_0$$

$\lambda_1 < \lambda_0$, $\lambda_2 > \lambda_0$

Photoemission

No emission of Photon